Figure 1: Before and After Images

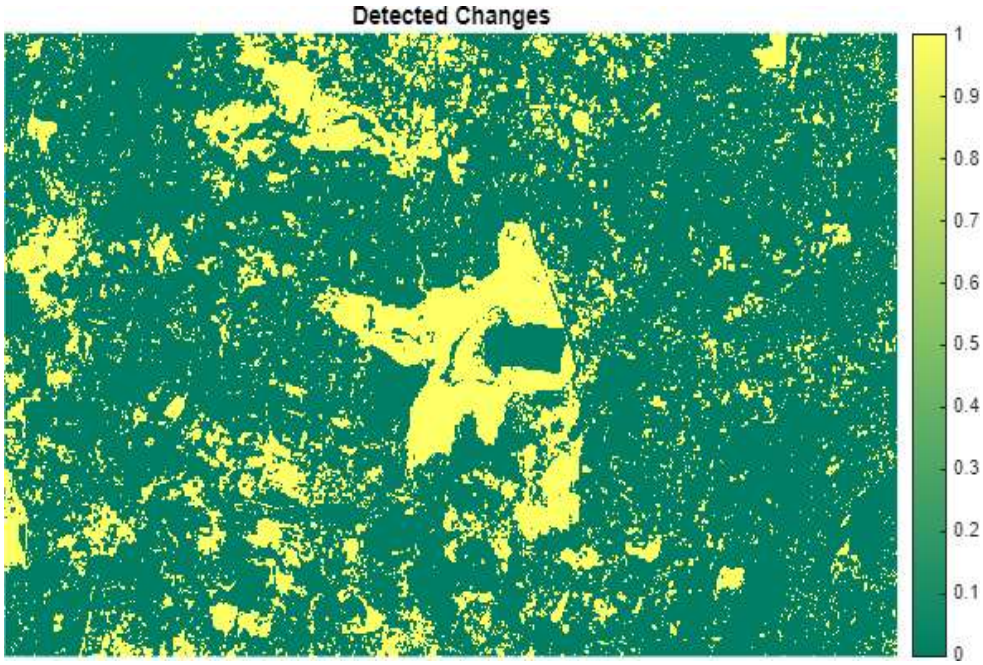

Figure 2: Detected Changes

```
Total Deforestation: 11.36%
Status: Deforestation has increased over the years.
Deforestation Percentage Over the Years:
    Year    Deforestation_Percent
    ____    _____________________

    2000                 0
    2005            5.4886
    2010            9.5124
    2015            13.074
    2020            16.168
    2025            18.876
```

Figure 3.1: Deforestation Percentage Rate

```
Predicted Deforestation Loss for Future Years:
    Year    Predicted_Deforestation_Loss
    ____    ____________________________

    2025                18.755
    2030                20.581
    2035                21.778
    2040                22.345
    2045                22.283
    2050                21.592
```

Figure 3.2: Predicted Deforestation Rate

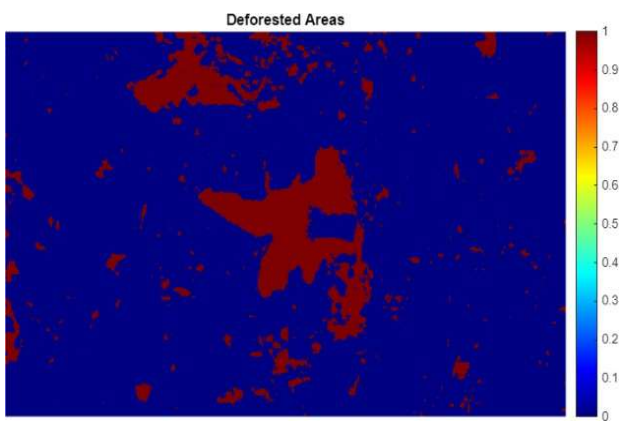

Figure 3.3: Deforested Areas Overlay

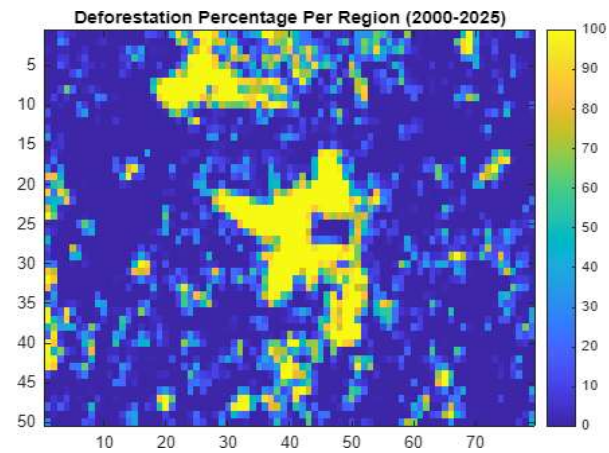

Figure 3.4: Deforestation % per Region

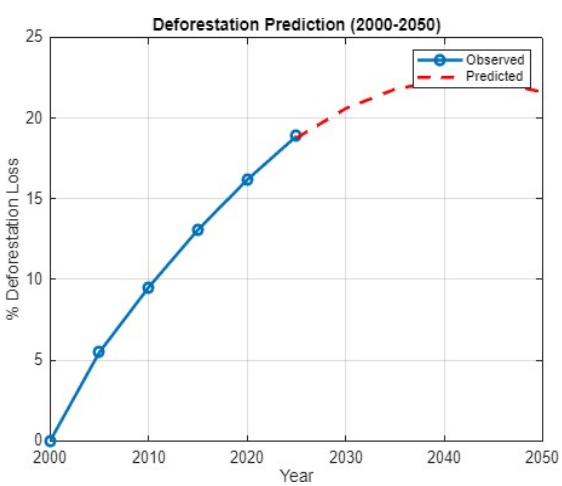

Figure 3.5: Predicted Deforestation Trend

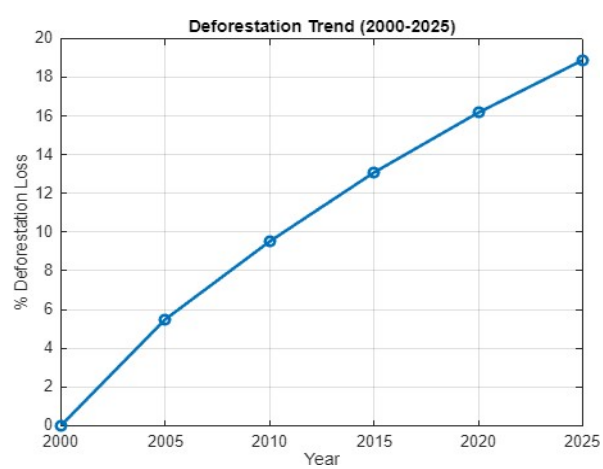

Figure 3.6: Observed Deforestation Trend

```
Water Area in 2000: 14.48%
Water Area in 2025: 7.77%
Total Water Body Reduction: 46.31%
Status: Reduction in Water Body
```

Figure 4.1: Water Body Change Rate

```
Predicted Water Body Change for Future Years:
    Year    Predicted_Water_Change
    ____    ______________________

    2025            7.7737
    2030            6.4327
    2035            5.0916
    2040            3.7506
    2045            2.4095
    2050            1.0684
```

Figure 4.2: Predicted Water Change Rate

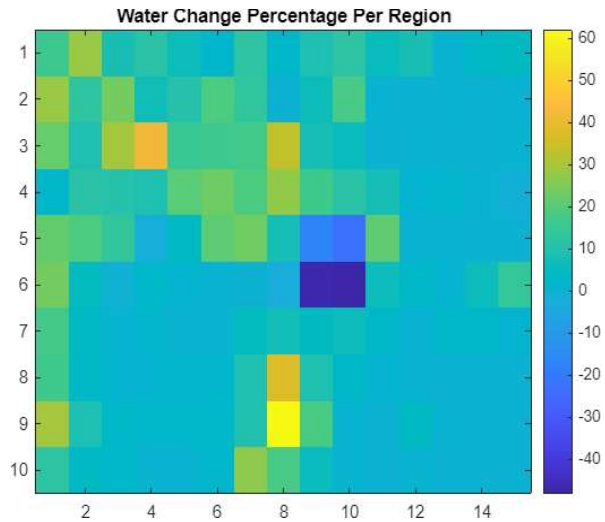

Figure 4.3: Water Change % per Region

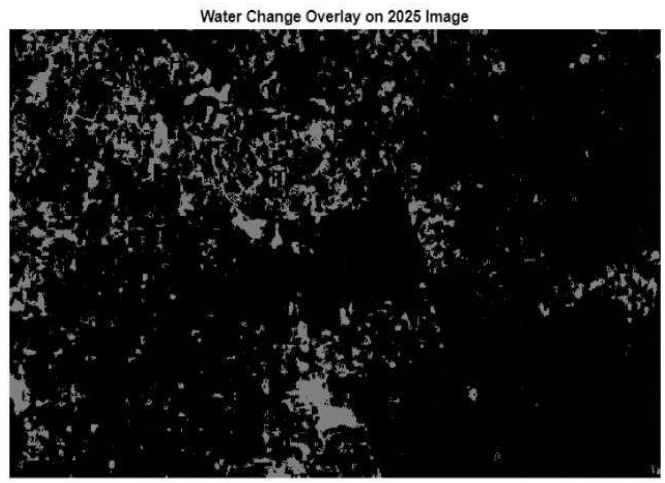

Figure 4.4: Water Body Change Overlay

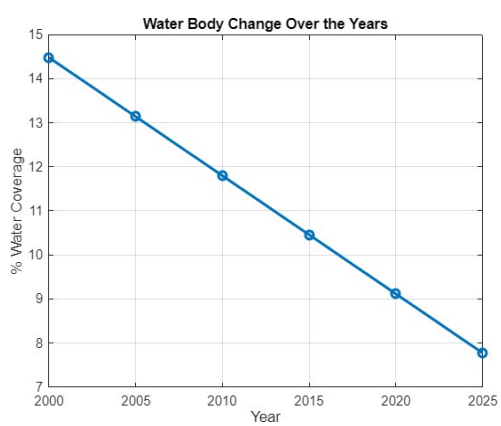

Figure 4.5: Observed Water Change Trend

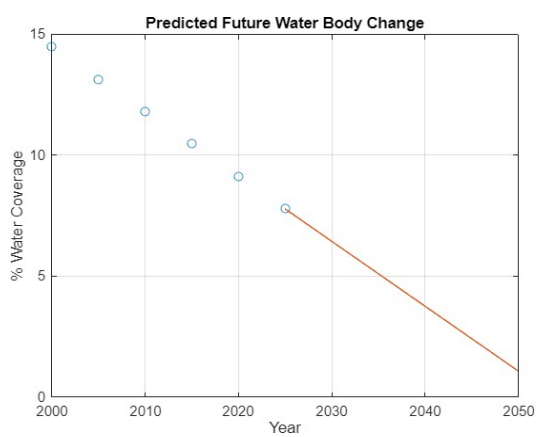

Figure 4.6: Predicted Water Change Trend

```
Total Urbanization Growth: 0.83%
Status: Urbanization has increased over the years.
Urbanization Percentage Over the Years:
```

Year	Urbanization_Percent
2000	0
2005	0.045707
2010	0.49571
2015	1.1924
2020	2.0944
2025	3.1694

Figure 5.1: Urbanization Percentage Rate

```
Predicted Urbanization for Future Years:
```

Year	Predicted_Urbanization
2025	3.201
2030	4.5944
2035	6.2362
2040	8.1264
2045	10.265
2050	12.652

Figure 5.2: Predicted Urbanization Rate

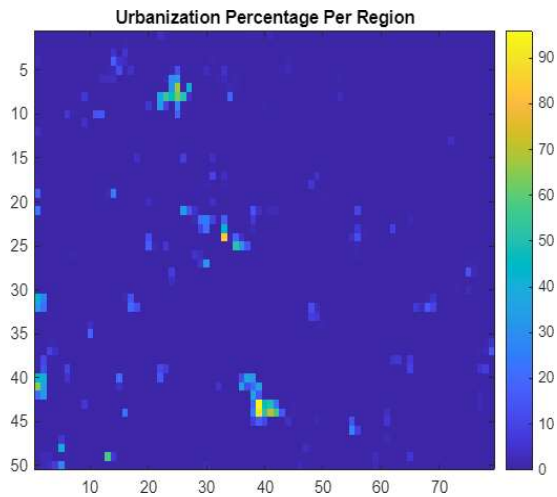

Figure 5.3: Urbanization % per Region

Figure 5.4: Urbanization Overlay

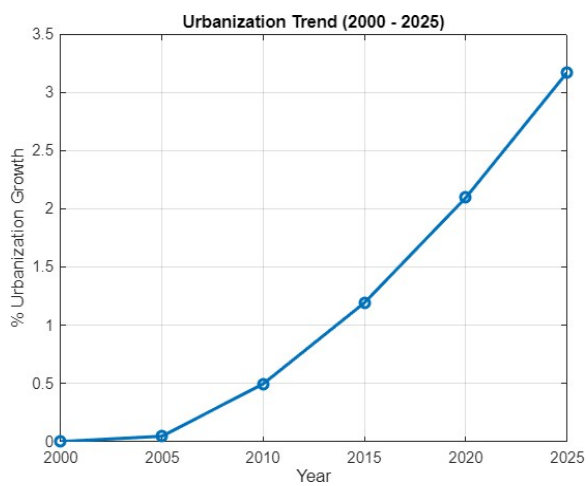

Figure 5.5: Observed Urbanization Trend

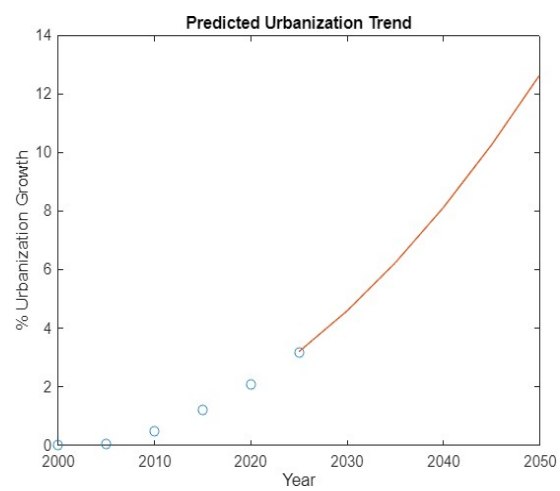

Figure 5.6: Predicted Urbanization Trend

```
Shoreline Coverage in 2000: 14.48%
Shoreline Coverage in 2025: 7.77%
Total Shoreline Retreat (Sea Level Rise): 46.31%
Status: Shoreline Retreat (Sea Level Rise)
```

Figure 6.1: Shoreline Retreat Percentage Rate

```
Predicted Shoreline Change for Future Years:
    Year    Predicted_Shoreline_Change
    ____    __________________________

    2025              7.7737
    2030              6.4327
    2035              5.0916
    2040              3.7506
    2045              2.4095
    2050              1.0684
```

Figure 6.2: Predicted Shoreline Retreat Rate

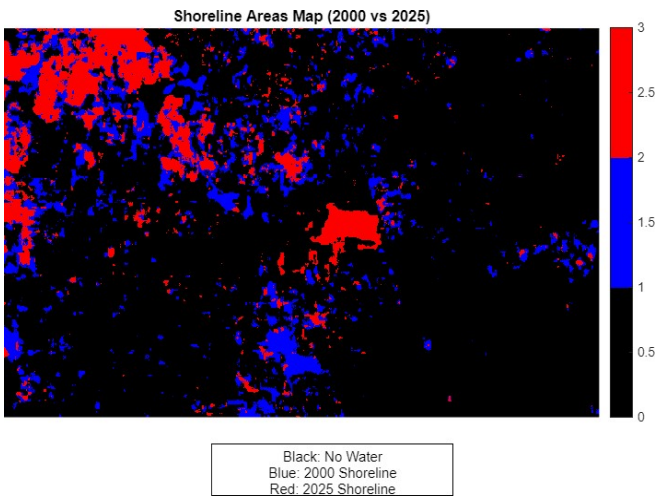

Figure 6.3: Shoreline Area Map

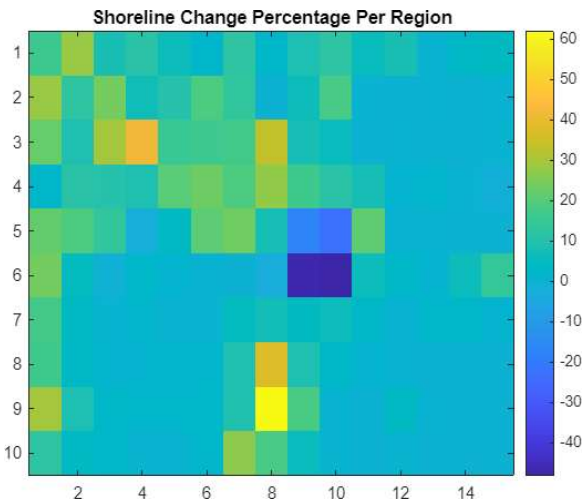

Figure 6.4: Shoreline Retreat % per Region

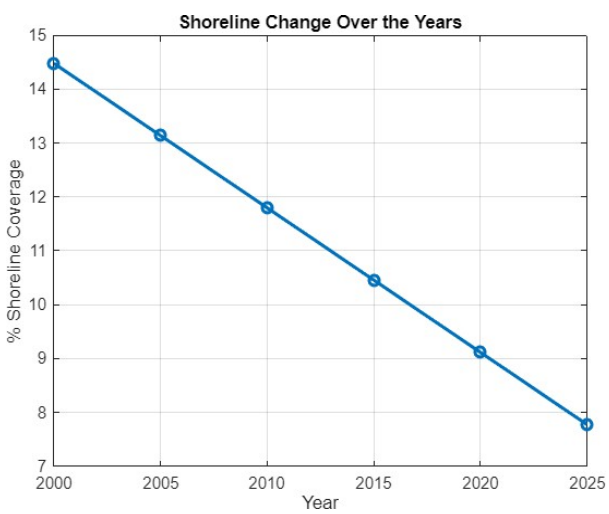

Figure 6.5: Observed Shoreline Change Trend

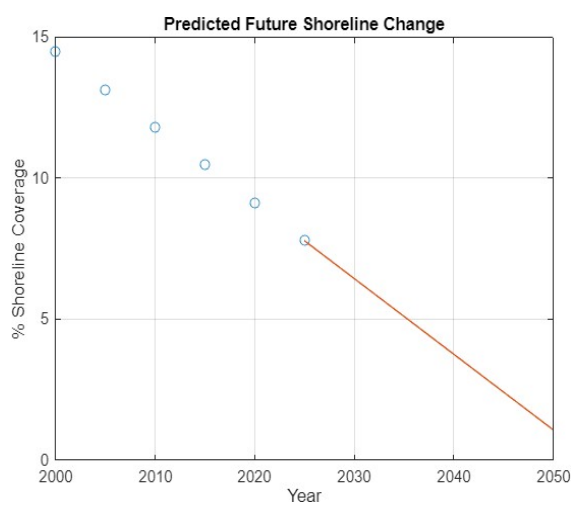

Figure 6.6: Predicted Shoreline Change Trend

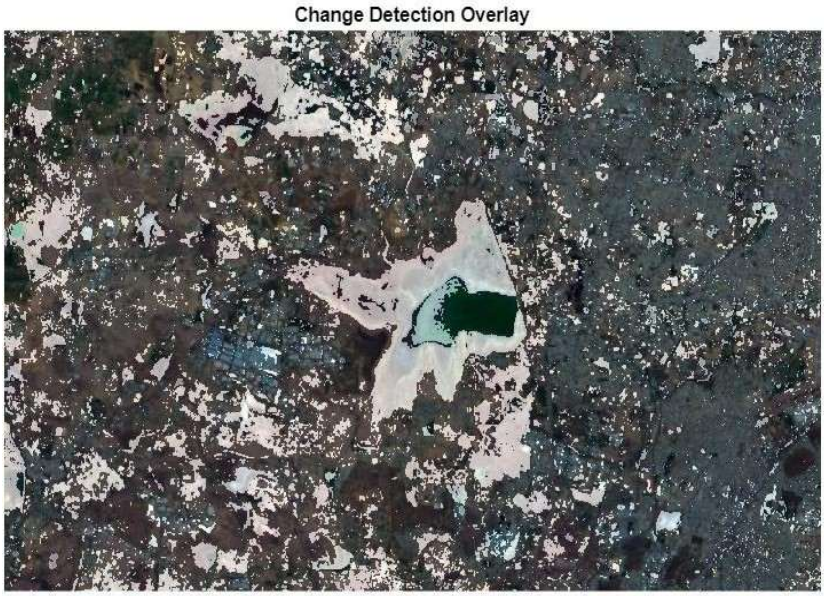

Figure 7: Change Detection Overlay

Figure 8: Registered Image Comparison